Natural Language Processing & Machine Learning

Roadmap

ML → Supervised → CLASSIFICATION
ML → Supervised → REGRESSION
ML → Unsupervised

Model

Dependent Feature → Prediction

I/P Independent features ← f_1 f_2 f_3 f_4

O/P

Text

Spam Classification → Email → Spam / No spam

Email Subject → Billionaire

Email Body → "You won a lottery of Billion $$"

Spam/Ham

1 ⇒ Natural Language Processing

Meaningful vector

① Alexa
② Google Home

Roadmap of NLP

NLTK, Spacy

Tensorflow/ Pytorch

BERT

TRANSFORMER

Text Preprocessing → Word Embedding

Neural N/w

RNN, LSTM RNN, GRU RNN → Deep Learning

I/P Text → vector

Text Preprocessing 3 → Word2Vec, Avgword2Vec

Input Text → vector

Text Preprocessing 2 → BOW, TFIDF, Unigrams, Bigrams

Cleaning the input

Text Preprocessing → Tokenization, Lemmatization, Stemming, Stopwords

Python ✓

↳ Machine Learning

Note: Tensorflow and Pytorch both are open source.

- Tensorflow is made by Google.
- Pytorch is made by Facebook.